Unit 20 Triggering for Improved Sensitivity: ~~Difficulty~~

(1) Examples of Triggering 209

(1.) What is Triggering in Online Experiment?

Triggering is a mechanism to decide when a user becomes part of an experiment. Instead of tracking everyone who visits a website, we only measure users who do something relevant (e.g. click, search, etc).

#section: Examples of Triggering (page 209)

This section gives real-world examples of triggering conditions that improve experiment sensitivity by focusing only on relevant ~~(e.g. click, search, etc)~~ users:

Here are the core examples they list and the intuition behind each:

(2) Examples of Triggering

1. Search Engine Triggering

- Trigger condition: A user perform a search.
- Why? You're testing changes to the ranking algorithm or search ~~AI~~. UI.
- Don't include users who never search — they're irrelevant noise.

> Real-World Analogy
>
> Testing a new spellchecker only for people who type in words. Don't include those who just open it and read documents

2. Ad Click Triggering

- Trigger Condition: User clicks on an ad.
- Why? The treatment affects landing page experience
- You shouldn't include users who never saw or clicked the ad — they dilute the effect.

> Real-World Analogy
> You're testing a new store layout. Only consider customers who walked into the store, not those who passed by on the street.

3. Shopping Cart Triggering

- Trigger Condition: User adds an item to the cart.
- Why? You're testing a new checkout flow or discount display.
- Users who never intended to buy (browsers) shouldn't be included.

> Real-world Analogy:
> Testing a new cashier line system. Only evaluate users who actually want to the checkout line.

4. Video Watch Triggering

- Trigger Condition: User clicks "Play" on a video.
- Why? You're testing a new buffering algorithm or subtitle UI
- User who don't watch video are irrelevant.

> Real-world Analogy
> Testing noise-cancelling headphones only matters for people who turn them on, not people who

(3) Why Use Triggering? (Ties to statistical Power)

No triggering (all-user inclusions)

You get higher sample size, but also much more noise and lower sensitivity. The signal-to-noise ratio worsens.

With triggering:
You reduce irrelevant users, leading to:

- ↑ Increased sensitivity
- ↑↑ Smaller MDEs (Minimum Detectable Effect)
- ↑↑ Faster conclusions

(4) Math Behind Triggering (Simple Derivation)

Let's define:

- N: Total number of users in experiment.
- N_T: Number of triggered users (e.g. those who search or add to cart)
- p: Probability of a user triggering.
- μ_C, μ_T: Mean metric value in control/Treatment.
- σ^2: Variance.

#Estimating Sensitivity

The Standard Error (SE) of a metric is:

$$SE = \frac{\sigma}{\sqrt{N}}$$

But if on $_pN$ user are relevant:

$$S.E._{triggered} = \frac{\sigma}{\sqrt{pN}}$$

This might look worse (since fewer users), but:

- The variance σ^2 among triggered users is much lower (more homogeneous behavior)
- Also, the effect size is stronger among the triggered users.

Thus, triggered test often end up being more powerful despite using fewer samples.

(5) Concept Overlap with Other Topics.

Concept	Link to Triggering
Variance Reduction (Unit 18)	Triggering reduces between-user variance by filtering only engaged users.
Sensitivity & Power (Unit 17)	Fewer users but stronger signal increases power.
Pitfalls (Unit 20.6)	Biased triggering (e.g. using post-treatment behavior) leads to invalid results
Overall Treatment Effect (Unit 20.4)	Need to generalize triggered results carefully to overall population.

(8) Summary.

Concept — Triggering

Purpose → Increase sensitivity by focusing on relevant user behaviour.

Mechanism → Include only users who take a predefined actions

Benefit → Better power, smaller variance, faster insights.

Risk → Must trigger before seeing treatment to avoid bias.

Math → Reduces variance, increases signal-to-noise

Analogy → like testing a feature only on those who use it, not everyone.

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(2) A Numerical Example (Kohavi, Longbotham et al. 2009) 212

(1) Purpose of This Section

This numerical example demonstrates how triggering can significantly improve sensitivity in A/B tests — using real-world data simulation from Bing search engine experiments.

It shows:

- How mean differences can be drowned out when untriggered (irrelevant) users are included.
- How filtering only triggered users (those affected by treatment) amplifies treatment effects and improves statistical power.

(2) The Numerical Example (Reconstructed in Simple Terms)

Imagine a scenario at Bing:

- A new feature is designed to change the layout of search results.
- It only affects users who perform a search (trigger condition).
- But if we analyze all users, including those who don't search, the treatment effect gets buried.

User Data

Let's say we have 10,000 users.

User Type	% of Users	Metric (Page load time)
Triggered Users (Performed search)	40% (4000 users)	Treatment increases page load by ↑0.2 sec.
Untriggered Users (No search)	60% (6000 users)	No effect (same for control & treatment)

True Effect

- Control Group : No change
- Treatment Group (Triggered Only) : +0.2 second.
- Overall Treatment Group (All users) : Effect is diluted

So, the true effect among triggered users is :

$$\Delta_{triggered} = +0.2 \text{ sec}$$

But in the overall user base

$$\Delta_{overall} = 0.4 \times 0.2 + 0.6 \times 0 = 0.08 \text{ sec}.$$

This small average effect (0.08 sec) is harder to detect due to variance/noise.

Real-World Analogy

You're testing a new checkout interface in a store.

- Only people who go to the cashier see the change
- But if you compare all visitors, including window shoppers, the overall effect gets blurred.

Why This Matters

Metric Type	Value
True Effect (Triggered Users)	+0.2 sec
Observed Overall Effect (All users)	+0.08 sec
Variance (All users)	Higher (more noise)
Statistical Power	Lower (without triggering)

So:

- Triggering gives you a more accurate view of actual treatment impact.
- Not triggering → diluted effect → harder to detect → false negatives.

(3) The Numerical setup (Based on the Book) / Another Example.

Let's say:

- You ~~can~~ run an experiment on search page behavior
- Only 20% of total users ever perform a search.
- The treatment only affects the search experience
- You're measuring revenue per user.

key Variables

Variable	Value (Example).
Numbers of users(N)	1,000,000.
Users who perform search(p)	20% or 0.2
Mean revenue (control, all users)	$0.60
Treatment effect (only on search users)	+$0.10
Baseline revenue (search users)	$2.00
std dev(search users)	$4.00
std dev (all users)	$5.00

Strategies

(A) Global Analysis (All users)

You're comparing mean revenue per user across control vs. treatment groups, even though only 20% are affected.

Observed Treatment Effect = 0.2 × 0.10 = $0.02

$$SE = \frac{5}{\sqrt{1{,}000{,}000}} = 0.005$$

$$z = \frac{0.02}{0.005} = 4.0 \Rightarrow \text{significant.}$$

☑ This seems okay — you can detect effect.

(B) Triggered Analysis (only searchers).

Now we analyze only the 20% users (200,000) who triggered a search — the ones who are actually affected.

$$\text{Effect} = \$0.10, \quad SE = \frac{4}{\sqrt{200{,}000}} = 0.0089$$

$$z = \frac{0.10}{0.0089} = 11.2 \Rightarrow \text{Highly significant}$$

Triggered analysis is very more sensitive, even with fewer users because:

- It isolates only relevant population
- It removes irrelevant noise.

Math! Why Triggering Works.

Variance Comparisons:

Assume:

- All users: higher variance due to mixed behaviors
- Triggered users: more homogeneous, lower variance.

let:

- $\sigma_{all} = 6$
- $\sigma_{triggered} = 4$
- $N = 1,000,000$
- $N_{triggered} = 200,000$

Even though $N_{triggered} < N$, the effect size is stronger and variance smaller in triggered group. So Standard Error(SE) becomes better for detecting true treatment effects.

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(3) Optimal and Conservative Triggering (Page 213)

X 1) What's The's Section About? X

→ This section addresses how to choose the triggering condition - and the trade-off between.

☑ Optimal triggering : Maximizes statistical power, detects effects faster.

☐ Conservative triggering : Avoids bias, keeps experiments trustworthy.

> The core idea: Triggering must balance sensitivity and correctness.

X 2. The Two Styles. X

1. Optimal Triggering (Focus only on affected users to maximize power)

Goal: Improve experiment sensitivity - i.e. make it easier to detect small effects.

How:

Trigger on only the users most likely ~~to~~ to:

- be affected by the treatment.
- exhibit strong treatment effect.
- have low variance in their behavior.

This improves:

- ☑ Effect size (larger signal)
- ☑ Reduced variance (less noise)
- ☑ Statistical power (higher z-score with fewer samples)

Real-world analogy
Want to test a new athletic shoe?
Trigger only runners, not everyone in the mall.

2. Conservative Triggering
(use only pre-treatment criteria to ensure trustworthiness)

Goal: Avoid bias and ensure valid inference

How:
Trigger based on pre-treatment behavior, or external user properties that are independent of treatment, exposure.

This avoids:
- ✗ Biased sample selection.
- ✗ Post-treatment leakage (treatments causing triggering),
- ✗ inflated false positives.

Real-world analogy
If you're testing a diet plan, don't include people only after they start losing weight (that is post-treatment bias)

The Trade-off (key-Message)

Triggering style	Pros	Cons
☑ Optimal Triggering	High sensitivity, faster	Risk of bias if not careful.
◉ Conservative	Unbiased, trustworthy	May include irrelevant user (more noise).

X 3. Math Insight X.

Let:

- $\mu_T - \mu_C$: True treatment effect
- σ^2: variance
- n: sample size

Your statistical power (ability to detect an effect) is:

$$z = \frac{\mu_T - \mu_C}{\sigma / \sqrt{n}}$$

Optimized triggering improves $\mu_T - \mu_C$ (more relevant users) and reduces σ (more consistent users), which increases z-power.

But if the triggering condition depends on treatment, the sample is biased, and the estimate is no longer valid.

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④ Overall Treatment Effect 2'4

④ What is the "Overall Treatment Effect"?

This section addresses a key statistical challenge:

> How do you estimate the true effect of a treatment on the entire population, when your analysis only include triggered users?

Triggering improves sensitivity, but it hides the overall impact unless adjusted properly.

Problem setup:

You're running an experiment where:

- Only 20% of users trigger (e.g. search, click, purchase)
- You observe a strong treatment effect among triggered users.
- But you need to estimate:
  => "What is the average treatment effect across all users?"

Core Idea:

> The Overall Treatment Effect is a weighted combination of:
> - "Triggered group effect"
> - "Non-triggered group effect (usually zero)"

Formula

$$\text{Overall Effect} = p \cdot \Delta_T + (1-p) \cdot \Delta_N$$

where:

- p: Proportion of user who triggered.
- Δ_T: Treatment effect among triggered users.
- Δ_N: Treatment effect among non-triggered users (often 0 if treatment doesn't affect them).

In most triggering setups:

- $\Delta_N \approx 0$ (user who didn't trigger are unaffected)

So this simplifies to:

$$\text{Overall Effect} \approx p \cdot \Delta_T$$

Real-world Analogy

Let's say you launch a new discount coupon, but it's only visible to users who add items to their cart.

- You find that cart users spend $5 more in treatment groups.
- But only 25% of users add items to cart.

So your Overall Effect across all users is:

$$0.25 \times 5 = \$1.25$$

So even though the treatment worked very well for some, its global business impact is smaller.

② Key Assumptions (And Pitfalls if violated).

Assumption	Why it Matters
No Post-Treatment Triggering	Triggering must be based on pre-treatment behavior. if the treatment itself causes triggering, the estimate is biased.
✗ No Hidden Spillovers	If treatment somehow affects non-triggered users, assuming $\Delta_N = 0$ is invalid.
☑ Random Assignment	Randomization ensures triggered proportions are valid across group.

③ Summary

Concept	Explanation
What	→ The overall treatment effect is the average treatment impact across all users, not just those who triggered.
Formula	→ $\Delta = p \cdot \Delta_T + (1-p) \cdot \Delta_N$
Gotcha	→ If treatment causes triggering, then Δ_T is biased and so is overall effect.
Why it Matters	→ Business stakeholders need ~~to~~ the total impact, not just the effect on a subgroups.

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(5) Trustworthy Triggering (Page 215).

(1) What is Trustworthy Triggering?

→ This sections focus on a critical requirement in experimentation:

✓ Triggering must be done in a way that preserves the statistical validity of the experiment.

✗ Never allow treatment to influence whether a user is triggered.

This is non-negotiable for a trustworthy experiment.

The Cona Principle: Triggering Must be Independent of Treatment.

This means:

> A user's inclusion in the analysis group (triggering) must NOT be influenced by the treatment they received.

Let's break this down with a real-world story and then get to the math.

Analogy: ~~Tractor~~ New search layout

Let's test a new search results layout

- You assign user randomly to control or treatment.
- You trigger analysis only when a user perform a search.

Now, two scenarios:

☑ Case A : Triggering is independent of Treatment

Trigger = "user performs a search"

☑ This is safe, because:

- users decide to search before seeing search results (treatment)
- Treatment doesn't affect whether they search.
- ☑ This is trustworthy triggering.

✗ Case B : Triggering is caused by Treatment.

Imagine your new layout has a big glowing search Bar.

Now:

- Users in the treatment group are more likely to search
- You still trigger only when a user performs a search

But now:

- More treatment users get triggered than control
- Your treatment group include more engaged or curious users.

So:

| ⓘ You're comparing an engaged treatment sample vs. a less engaged control sample

⇒ That is untrustworthy, because treatment affected who got triggered.

② Relationship between Triggered Users and Treatment.

First, define both:

- Treatment group: Users who receive the experimental version (e.g. new button, new layout)
- Triggered users: Subset of users who did something specific that makes them relevant for analysis (e.g. performed ~~a search, visited~~ a search, visited product page)

Key Relationship Rule

> ⚠ The triggering condition must ~~NOT~~ be influenced by the treatment.

This means:

- Triggering is independent of whether the user is in the treatment or control group.
- Every user should have equal chance to be triggered, regardless of which group they're in.

Why?

Because if treatment affects whether a user gets triggered, then:

- You're not analyzing the same kind of users in both groups.
- Your results become biased.
- It violates the core principle of randomized controlled experiments.

Summary Table

Question	Trustworthy.
Does treatment decide who gets triggered?	✗
Is triggering based only on pre-treatment behavior	✓
And both groups triggered by same rule, regardless of what they see?	Yes

Final Takeaway.

- ☑ Triggering defines who is analyzed.
- ☑ Treatment defines what version they experience
- If treatment affects triggering, then your analysis is invalid.

(5) What happens if you violate this?

If treatment influences whether a user triggers, it causes:

- selection bias
- Invalid p-values
- Inflated treatment effects.
- wrong business decisions

This is a core experimental flaw:

(4) Summary

Concept	Explanation
Trustworthy Triggering	Triggering must not depend on the treatment
why?	To preserve validity of the experiment and avoid selection bias
pitfall	Post-treatment behavior (like clicking a new UI element) must not be used for triggering
Rule	Use only pre-treatment data or randomization for triggering logic.
outcome	Ensures the treatment effect estimate is unbiased and generalizable.

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(6) Common Pitfalls Page 216.

(1) what is this section about?

→ This sections list frequents mistakes (pitfalls) people make when using triggered analysis in experiments.

These mistakes often lead to:

- ✗ Biased results
- ✗ invalid p-values
- ✗ Misleading treatment effects
- ✗ Wrong product or business decisions.

These pitfalls are mostly about misusing or misunderstanding triggering.

Common Pitfall #1: Triggering Depends on Treatment

↳ (5) Treatment only Triggering.

** Pre-vs Post- Treatment Triggering **

Concept	Pre-Treatment	Post-Treatment Triggering
When the trigger happens	Before the user sees the treatment	After the user has seen the treatment.
Fairness	☑ Fair: Same rules apply to both groups.	☒ Unfair: Rules depend on treatment exposure.
What it captures	who should be analyzed based on neutral behavior	who reacted to the treatment.

	☑	☒
Validity	Maintain randomization and unbiased estimates	Introduce selection bias and invalid causal inference.
Example	"User visited homepage" (before seeing new layout)	"User clicked new button" (only available in treatment).

Analogy.

Type	Example	What happens.
☑ Pre-treatment Triggering	Analyze all visitors who walks into the restaurant.	Fair comparison — same entry condition
☒ Post-treatment Triggering	Analyze only users who ordered the burger	Biased — treatment banner influences who you analyze.

Common Pitfall #2 : Post-Treatment Behavior in Triggering.

> Triggering condition includes behavior that happened after the treatment was applied.

This causes leakage — you're conditioning on a consequence of the treatment.

> Example: "only include users who searched and clicked the new UI."

But clicking the new UI was caused by treatment, so you're again selecting a non-random sample.

~~Comm~~. Common Pitfall #3 : Misinterpreting Overall Treatment Effect

You trigger on a subgroup (e.g. searches), see a big effect, and assume it applies to all users.

But if:

- only 20% triggered.
- Treatment effect £5 in this group

Then:

- The overall effect = 0.2 × £5 = £1

Not adjusting for this will overstate the impact.

This overlaps with:

→ Unit 20.4: Overall Treatment Effect (page 214).

Common Pitfall #4 : Ignoring Differential Triggering Rates.

If your triggering condition

- is "equally likely" in control and treatment → ☑ safe.
- is more likely in treatment (because of the treatment) → ✗ Biased.

This difference leads to sampling bias

overlaps with

→ Unit 20.5 : Trustworthy Triggering (page 215).

(2) Root Cause of All Pitfalls

> They all stem from violating the assumption of random sampling is statistical inferences.

When treatment affects who enters your analysis, you no longer have comparable groups, so your A/B test becomes invalid.

(3) Fixes and Best Practices

Problem	Fix
Treatment affects triggering	→ Define triggering before treatment exposure.
Triggering based on clicks/actions	→ Use pre-treatment behavior or random users.
Misinterpreting partial effect	→ Estimate overall effect using $p \cdot \Delta T$
Different trigger rules	→ Monitor trigger rate across arms and flag differences.

(4) Summary Table

Pitfall	Why it's bad	Fix
Trigger depends on treatment	Selection bias, inflated effect	Use pre-treatment criteria.
Trigger on post-treatment behavior	Violates causal inference	Never trigger on actions caused by treatment.
Misreading triggered effect as overall	Overstates business impact.	Use weighted adjustment: $p \cdot \Delta T$

Unequal trigger rates	skewed samples, invalid comparison	Compare trigger rates before analyzing.

(5) Final Takeaway:

- Triggered analysis is powerful - but dangerous if misused.
- Every one of these pitfalls leads to be false learning, which can harm product and business decisions.

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